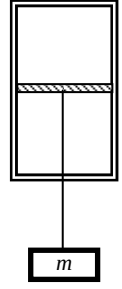


University Physics Quiz 5

Name: _____ SID: _____ School/Department: _____

1. A frictionless piston of negligible mass and heat capacity divides a vertical insulating cylinder of height $2h$ into two halves. Each half contains 1 mole of ideal gas at standard pressure and temperature. The heat capacity ratio $\lambda = c_p/c_v$ of the gas is given. A load of mass m is tied to the piston and suddenly released.



After some oscillations the system comes to equilibrium with the piston at rest and equal temperatures of the gases. What is the final displacement Δh of the piston assuming m is very large?

Solution: The conservation of energy gives

$$mg\Delta h = 2c_v(T_f - T_0) \quad (1)$$

where T_0 and T_f are the initial and final temperatures of the system respectively and c_v is the molar isochoric heat capacity of the ideal gas. The mechanical equilibrium requires

$$p_1 A - p_2 A - mg = 0 \quad (2)$$

where p_1 and p_2 are pressures in the lower and upper parts of the cylinder and A is the area of the piston. From equation of state of the ideal gas, we have

$$p_1 = \frac{RT_f}{A(h - \Delta h)}, \quad p_2 = \frac{RT_f}{A(h + \Delta h)}.$$

Substituting above equations into (2), we obtain

$$mg = RT_f \left(\frac{1}{h - \Delta h} - \frac{1}{h + \Delta h} \right) = \frac{2RT_f \Delta h}{h^2 - (\Delta h)^2}. \quad (3)$$

Combining (1) and (3), we obtain

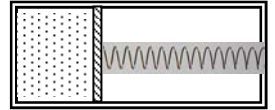
$$\frac{2c_v(T_f - T_0)}{\Delta h} = \frac{2RT_f \Delta h}{h^2 - (\Delta h)^2}. \quad (4)$$

If m is very large, from (1), we have $T_f \gg T_0$. Then, neglecting T_0 in (4) we have

$$\frac{c_v}{R} = \frac{(\Delta h)^2}{h^2 - (\Delta h)^2} \quad \text{or} \quad \frac{1}{\gamma - 1} = \frac{(\Delta h)^2}{h^2 - (\Delta h)^2}. \quad \left(\frac{c_v}{R} = \frac{c_v}{c_p - c_v} = \frac{1}{c_p/c_v - 1} = \frac{1}{\gamma - 1} \right)$$

$$\text{Therefore } (\gamma - 1)(\Delta h)^2 = h^2 - (\Delta h)^2, \quad \gamma(\Delta h)^2 = h^2, \quad \Delta h = \frac{h}{\sqrt{\gamma}}.$$

2. In a horizontal cylinder with a piston, there is a monatomic ideal gas to the left of the piston and vacuum to the right of the piston (see the figure). The cylinder is thermally insulated and the spring which connects the piston with the right wall is initially relaxed. After the piston is released and equilibrium is reached, the volume of the gas is doubled. What is the ratio of the final temperature to the initial temperature? What is the ratio of the final pressure to the initial pressure? Assume that the heat capacities of the cylinder, piston and spring can be neglected.



Solution: The conservation of energy gives

$$\Delta U + \frac{1}{2} kx^2 = 0$$

where $\Delta U = C_v(T_f - T_i)$ and x is the displacement of the piston $x = \frac{V_f - V_i}{A}$ (A is the area of the piston). Then, we

obtain

$$C_V(T_f - T_i) = -\frac{k(V_f - V_i)^2}{2A^2} \quad (1)$$

The mechanical equilibrium requires

$$p_f A = kx \quad \text{or} \quad p_f = \frac{kx}{A} = \frac{k(V_f - V_i)}{A^2}.$$

Using the equation of state $p_f = \frac{nRT_f}{V_f}$, we have

$$\frac{nRT_f}{V_f} = \frac{k(V_f - V_i)}{A^2}. \quad (2)$$

(1)/(2):

$$\frac{C_V(T_f - T_i)V_f}{nRT_f} = -\frac{V_f - V_i}{2} \quad \text{or} \quad \frac{C_V(T_i - T_f)}{nRT_f} = \frac{V_f - V_i}{2V_f} \quad \text{or} \quad \frac{2C_V}{nR} \left(\frac{T_i}{T_f} - 1 \right) = 1 - \frac{V_i}{V_f}.$$

Thus

$$\frac{T_i}{T_f} = \frac{nR}{2C_V} \left(1 - \frac{V_i}{V_f} \right) + 1.$$

Substituting $V_i/V_f = 1/2$, $C_V = \frac{3}{2}nR$ into above equation, we obtain

$$\frac{T_i}{T_f} = \frac{1}{3} \left(1 - \frac{1}{2} \right) + 1 = \frac{7}{6}.$$

Therefore
$$\frac{p_i}{p_f} = \frac{\frac{nRT_i}{V_i}}{\frac{nRT_f}{V_f}} = \frac{T_i}{T_f} \frac{V_f}{V_i} = \frac{7}{3}.$$

3. Assume that the internal energy of a mole of Van der Waals gas is given by

$$U = U_0 + C_V T - \frac{a}{V}$$

where C_V is a constant.

(a) Find the adiabatic equation of the Van der Waals gas.

(b) A Carnot cycle for the Van der Waals gas operates between two reservoirs of temperature T_1 and T_2 . Obtain the efficiency of the cycle.

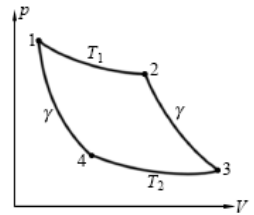
Solution: (a) From $U = U_0 + C_V T - \frac{a}{V}$,

we have
$$dU = C_V dT + \frac{a}{V^2} dV.$$

Using the first law, we have

$$dQ = dU + p dV = C_V dT + \frac{a}{V^2} dV + \left(\frac{RT}{V-b} - \frac{a}{V^2} \right) dV = C_V dT + \frac{RT}{V-b} dV$$

For a adiabatic process, $dQ = 0$. Then $C_V dT + \frac{RT}{V-b} dV = 0$ or $\frac{dT}{T} + \frac{R}{C_V} \frac{dV}{V-b} = 0$.



Thus

$$T(V-b)^{\frac{R}{C_v}} = C.$$

(b) For $1 \rightarrow 2$:

$$\Delta Q_{12} = \Delta W_{12} + (U_2 - U_1),$$

$$\Delta W_{12} = \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} \left(\frac{RT_1}{V-b} - \frac{a}{V^2} \right) dV = RT_1 \ln \frac{V_2-b}{V_1-b} + \left(\frac{a}{V_2} - \frac{a}{V_1} \right),$$

$$U_2 - U_1 = -\frac{a}{V_2} + \frac{a}{V_1},$$

$$\Delta Q_{12} = RT_1 \ln \frac{V_2-b}{V_1-b}.$$

For $3 \rightarrow 4$: $\Delta Q_{34} = \Delta W_{34} + (U_4 - U_3) = RT_2 \ln \frac{V_4-b}{V_3-b}.$

For a complete cycle, $\Delta U = 0$. By using the first law, we have

$$W = \Delta Q_{12} + \Delta Q_{34} = RT_1 \ln \frac{V_2-b}{V_1-b} - RT_2 \ln \frac{V_3-b}{V_4-b}.$$

The heat input for one cycle is

$$Q_{in} = \Delta Q_{12} = RT_1 \ln \frac{V_2-b}{V_1-b}.$$

Therefore the efficiency is

$$\eta = \frac{W}{Q_{in}} = \frac{T_1 \ln \frac{V_2-b}{V_1-b} - T_2 \ln \frac{V_3-b}{V_4-b}}{T_1 \ln \frac{V_2-b}{V_1-b}}.$$

Using the adiabatic equation, we have

$$\begin{aligned} T_1(V_2-b)^{\frac{R}{C_v}} &= T_2(V_3-b)^{\frac{R}{C_v}} \\ T_1(V_1-b)^{\frac{R}{C_v}} &= T_2(V_4-b)^{\frac{R}{C_v}}. \end{aligned}$$

Then we have

$$\frac{V_2-b}{V_1-b} = \frac{V_3-b}{V_4-b}.$$

Therefore, we have

$$\eta = \frac{W}{Q_{in}} = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}.$$

4. An ideal gas initially at a pressure p_0 undergoes a free expansion until its final volume is 8 times its initial volume V_0 .

The gas is then slowly and adiabatically compressed back to its initial volume at which the pressure is $2^{\frac{6}{5}} p_0$.

(a) Determine whether the ideal gas is monatomic, diatomic or polyatomic.

(b) Find the total change of internal energy in this process (in terms of p_0 and V_0).

*(c) Find the total change of entropy in this process.

Solution: (a) Since in the free expansion process, the internal energy of the ideal gas is unchanged, then, the temperature is also unchanged. From the equation of state, we know that the pressure of the gas after the expansion is $p_0/8$. Thus, in the adiabatic process, we have

$$\frac{p_0}{8}(8V_0)^\gamma = 2^{6/5} p_0 (V_0)^\gamma.$$

Therefore $8^{\gamma-1} = 2^{6/5} = 8^{2/5}$, $\gamma-1 = \frac{2}{5}$, $\gamma = \frac{7}{5}$, $C_V = \frac{nR}{\gamma-1} = \frac{5}{2}nR$ and the gas is diatomic.

(b) (Solution 1) Obviously, in the free expansion process, the work done to the system is zero. In the adiabatic compression process, the adiabatic equation is

$$pV^\gamma = C = \frac{p_0}{8}(8V_0)^\gamma = 2^{6/5} p_0 (V_0)^\gamma.$$

Therefore, the work done to the system is then

$$\begin{aligned} W &= - \int_{8V_0}^{V_0} p dV = - \int_{8V_0}^{V_0} \frac{C}{V^\gamma} dV = -2^{6/5} p_0 (V_0)^\gamma \int_{8V_0}^{V_0} \frac{1}{V^\gamma} dV = \frac{2^{6/5} p_0 (V_0)^\gamma}{\gamma-1} \frac{1}{V^{\gamma-1}} \Big|_{8V_0}^{V_0} \\ &= \frac{2^{6/5} p_0 (V_0)^\gamma}{\gamma-1} \left(1 - \frac{1}{8^{\gamma-1}}\right) \frac{1}{V_0^{\gamma-1}} = \frac{2^{6/5} p_0 V_0}{\gamma-1} \left(1 - \frac{1}{2^{6/5}}\right) = \frac{5 p_0 V_0}{2} (2^{6/5} - 1). \end{aligned}$$

Finally $\Delta U = W + Q = W = \frac{5 p_0 V_0}{2} (2^{6/5} - 1)$.

(Solution 2): $\Delta U = C_V (T_2 - T_1) = \frac{5}{2} nR (T_2 - T_1) = \frac{5}{2} (nRT_2 - nRT_1) = \frac{5}{2} (2^{6/5} p_0 V_0 - p_0 V_0) = \frac{5 p_0 V_0}{2} (2^{6/5} - 1)$.

(c) The change of entropy in the adiabatic process is zero. Then, making a reversible isothermal process connecting the states p_0, V_0 and $p_0/8, 8V_0$, we obtain

$$\Delta S = \int_{V_0}^{8V_0} \frac{dQ}{T} = \int_{V_0}^{8V_0} \frac{-dW}{T} = \int_{V_0}^{8V_0} \frac{p dV}{T} = \int_{V_0}^{8V_0} \frac{nR dV}{V} = nR \ln 8.$$

(In an isothermal process of an ideal gas, $dU = 0$.)